Deep Reinforcement Learning

From Zero to Hero

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Abstract

Lecture notes from for quick revision to RL.

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1

Preliminaries

Reinforcement learning draws on several areas of mathematics. This chapter collects the background used throughout the book: probability (Section 1.1), linear algebra (Section 1.2), calculus (Section 1.3) and information theory (Section 1.4). Readers comfortable with these topics may skip ahead and return here as needed.

1.1 Probability

We cover probability with a light touch on measure theory; these notes were prepared while following the lecture series Probability Foundations for Electrical Engineers (IIT M).

1.1.1 Introduction

Probability is the science of randomness that seeks to quantify patterns underlying seemingly non-deterministic events. Andrey Kolmogorov is regarded as the father of modern probability theory. We study probability rigorously because intuitive reasoning can often lead to paradoxes. A famous example is Bertrand's paradox, where different definitions of "randomness" lead to different probability values. The problem is stated as follows: A chord is drawn at random in a circle. What is the probability that the chord is longer than the side of the equilateral triangle inscribed in the circle? This paradox arises because the sample space is continuous (infinite) and must therefore be defined carefully to avoid ambiguity.

1.1.2 Cardinality and Countability

Functions and Mappings.

Given two sets A and B, a *function* $f : A \to B$ is a rule that associates every element of A with a unique element of B. The set A is called the *domain*, and the set B is called the *codomain*. The *range* (or image) of f is the subset of B consisting of all elements that are actually mapped to by elements of A, i.e.,

$$\text{Range}(f) = \{\, f(a) \mid a \in A \,\} \subseteq B.$$

Note that every element of A must be mapped to exactly one element of B, although some elements of B may not receive a mapping from A.

Injective (One-to-One) Function. A function $f : A \to B$ is said to be *injective* if distinct elements of A map to distinct elements of B. Formally,

$$f(a_1) = f(a_2) \implies a_1 = a_2, \quad \forall a_1, a_2 \in A.$$

Equivalently, every element of the range $\text{Range}(f) \subseteq B$ has a unique preimage in A.

Surjective (Onto) Function. A function $f : A \to B$ is *surjective* if every element of the codomain B has at least one preimage in A; that is,

$$\forall b \in B,\ \exists a \in A \text{ such that } f(a) = b.$$

Hence, for a surjective function, the range equals the codomain: $\text{Range}(f) = B$.

Bijective Function. A function $f : A \to B$ is *bijective* if it is both injective and surjective. In this case, each element of B has exactly one preimage in A, and f is said to be *invertible*. Its inverse $f^{-1} : B \to A$ is also a well-defined function satisfying

$$f^{-1}(f(a)) = a, \quad \forall a \in A, \qquad \text{and} \qquad f(f^{-1}(b)) = b, \quad \forall b \in B.$$

Cardinality

Cardinality is a measure of the "size" or number of elements in a set. For finite sets, this is simply the number of elements. However, comparing the sizes of infinite sets is more subtle. For instance, both the set of natural numbers $\mathbb{N}$ and the set of real numbers $\mathbb{R}$ are infinite, yet they are not of the same size. How, then, can we compare their sizes?

Cantor's Argument. Two sets have the same cardinality if there exists a bijective (one-to-one and onto) function between them.

Definition. Let A and B be two sets.

1. A and B are said to have the same cardinality, denoted $|A| = |B|$, if there exists a bijective function $f : A \to B$.
2. If there exists an injective (one-to-one) function $f : A \to B$ but no bijection, then B has greater or equal cardinality, i.e., $|B| \geq |A|$.
3. If there exists an injective but not surjective function $f : A \to B$, then $|B| > |A|$.

Theorem. The cardinality of the real numbers $\mathbb{R}$ is strictly greater than that of the natural numbers $\mathbb{N}$.

Proof Sketch. (Using Cantor's diagonal argument (the ideas to construct new real number by flipping the diagonal entry which will be different from all existing entries).) Assume, for contradiction, that there exists a bijection $f : \mathbb{N} \to (0, 1) \subset \mathbb{R}$, meaning every real number in $(0, 1)$ can be written as a decimal expansion and listed as $f(1), f(2), f(3), \ldots$. Construct a new real number r by altering the n-th digit of $f(n)$ (for instance, replace the n-th digit by 5 if it is not 5, otherwise by 4). By construction, r differs from every $f(n)$ in at least one digit, so r is not in the list. This contradicts the assumption that f is surjective. Therefore, no bijection exists, and we conclude that

$$|\mathbb{R}| > |\mathbb{N}|.$$

Hence, the set of real numbers is uncountably infinite, while the set of natural numbers is countably infinite.

Countability

Definition.

- A set E is said to be **countably infinite** if it is equi-cardinal with $\mathbb{N}$, i.e., there exists a bijection between E and $\mathbb{N}$.
- A set is said to be **countable** if it is either finite or countably infinite.

Examples.

- The set of even numbers is equi-cardinal with $\mathbb{N}$ (via $f(n) = 2n$).
- The set of integers $\mathbb{Z}$ is countably infinite (we can enumerate them as $0, 1, -1, 2, -2, \ldots$).
- The set of rational numbers $\mathbb{Q}$ is also countably infinite.

Theorem. The set of rational numbers $\mathbb{Q}$ is countably infinite.

Proof Sketch. Every rational number can be written as a fraction $\frac{p}{q}$ where $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. We can list all such pairs (p, q) in a two-dimensional grid indexed by integers p and positive integers q. By enumerating these pairs in a diagonal manner (e.g., by increasing $|p| + q$) and skipping equivalent fractions (such as $\frac{1}{2}$ and $\frac{2}{4}$), we can construct a sequence that lists all rational numbers exactly once. Thus, there exists a bijection between $\mathbb{Q}$ and $\mathbb{N}$, implying that $\mathbb{Q}$ is countably infinite.

$$|\mathbb{Q}| = |\mathbb{N}|.$$

Note. Uncountable sets (such as $\mathbb{R}$) are "larger" infinities than $\mathbb{N}$.

Theorem. Let I be a countable index set, and let $\{A_i : i \in I\}$ be a collection of countable sets. Then the union $\bigcup_{i \in I} A_i$ is countable. In other words, a countable union of countable sets is countable.

Proof Sketch. For each $i \in I$, enumerate the elements of A_i as $A_i = \{a_{i1}, a_{i2}, a_{i3}, \ldots\}$. Arrange these elements in a two-dimensional grid with rows indexed by i and columns by j. Since both I and each A_i are countable, the set of all pairs (i, j) is countable (using the same diagonal argument as before). Now define a mapping from $\mathbb{N}$ to $\bigcup_{i \in I} A_i$ by traversing this grid diagonally and listing each a_{ij} encountered. This provides an enumeration of the entire union, proving that

$$\left| \bigcup_{i \in I} A_i \right| = |\mathbb{N}|.$$

Hence, the countable union of countable sets is countable.

Examples of Countable Sets

1. **Algebraic Numbers:** The set of all real numbers that are roots of polynomials with integer coefficients. Since each polynomial has finitely many roots, and there are countably many such polynomials, the set of algebraic numbers is countable.
2. **Computable Numbers:** The set of all real numbers that can be computed to any desired degree of precision by a terminating algorithm or program. As there are only countably many possible programs, this set is also countable.

Uncountable Sets

Definition. A set F is said to be **uncountable** if it has cardinality strictly greater than that of $\mathbb{N}$, i.e., $|F| > |\mathbb{N}|$. Equivalently, there exists an injective (one-to-one) mapping from $\mathbb{N}$ to F, but no bijection between them.

Examples.

- The set of real numbers $\mathbb{R}$.
- The set of irrational numbers $\mathbb{R} \setminus \mathbb{Q}$.
- The set of all infinite binary strings $\{0, 1\}^\infty$.
- The power set of the natural numbers $\mathcal{P}(\mathbb{N})$.

Theorem. The set $\{0, 1\}^\infty$ of all infinite binary strings is uncountable.

Proof Sketch. (Sketch via Cantor's diagonal argument.) Assume that $\{0, 1\}^\infty$ is countable. Then all binary strings can be listed as:

$$s_1, s_2, s_3, \ldots$$

where each $s_i = (s_{i1}, s_{i2}, s_{i3}, \ldots)$ is an infinite sequence of 0s and 1s. Construct a new binary string $t = (t_1, t_2, t_3, \ldots)$ by flipping the i-th digit of s_i, i.e.,

$$t_i = \begin{cases} 0, & \text{if } s_{ii} = 1, \\ 1, & \text{if } s_{ii} = 0. \end{cases}$$

By construction, t differs from each s_i in at least one position (the i-th). Hence, t is not in the list, contradicting the assumption that the set is countable. Therefore, $\{0, 1\}^\infty$ is uncountable.

Remark. To show that another set A is uncountable, it suffices to construct an injective (or bijective) mapping from $\{0,1\}^\infty$ to A. For instance, the interval $[0,1]$ is uncountable because each real number in $[0,1]$ can be represented as an infinite binary sequence.

1.1.3 Probability Spaces

Random Experiment and Sample Space

Definition. A **random experiment** is a process that leads to one of several possible outcomes, where the outcome cannot be predicted with certainty in advance.

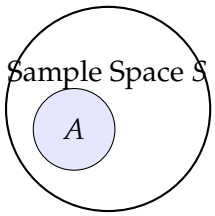
Sample Space. The **sample space** S (also denoted Ω) is the set of all possible outcomes of a random experiment that we are interested in:

$$S = \{\text{all possible outcomes}\}.$$

Example.

- Fair coin toss: coin faces or number of tumbles in air
- Velocity with which the coin strikes the ground $\mathbb{R}^+$

Remark. The definition of S depends on the level of detail we are interested in. For example, if we include physical parameters of a coin toss (e.g., velocity, angle, or spin), then S may be uncountable. An outcome is denoted by $w \in S$.



Event $A \subseteq S$

Events

Definition. An **event** is any subset of the sample space S which are of intereset. An event occurs if the outcome of the experiment belongs to that subset.

Example. When tossing two coins:

$$S = \{HH, HT, TH, TT\}.$$

Let A = "at least one head" $\Rightarrow A = \{HH, HT, TH\}$.

Note.

- Not all subsets of S are necessarily events (especially in uncountable S).
- Every event is a subset of S.
- In uncountable spaces, only measurable subsets are valid events.
- some subsets don't behave nicely, so we study it from the lens of measurable space.

Algebra and σ-Algebra of Events

Algebra of Sets. A collection $\mathcal{F}_0$ of subsets of S is called an **algebra** if:

1. $S \in \mathcal{F}_0$,
2. If $A \in \mathcal{F}_0$, then $A^c \in \mathcal{F}_0$,
3. If $A_1, A_2 \in \mathcal{F}_0$, then $A_1 \cup A_2 \in \mathcal{F}_0$.

That is, $\mathcal{F}_0$ is closed under complement and finite unions.

Example. Consider tossing a coin until the first head appears. Then $S = \{H, TH, TTH, TTTH, \ldots\}$, which is countably infinite. The algebra $\mathcal{F}_0$ does not necessarily include infinite unions of its sets.

σ-Algebra. A collection $\mathcal{F}$ of subsets of S is called a **σ-algebra** if:

1. $S \in \mathcal{F}$,
2. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$,
3. If $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

Note. Every σ-algebra is also an algebra, but the converse is not true. $\mathcal{F}$ is basically a subset of power-set of S.

The cardinality of $\mathcal{F}$ could be uncountable, it's matter of how big the uncountable is eg. $2^{\mathbb{N}}$ vs $2^{\mathbb{R}}$.

Measure and Probability Measure

Measurable Sets. Subsets belonging to $\mathcal{F}$ are called **$\mathcal{F}$-measurable sets**. The pair $(S, \mathcal{F})$ is called a **measurable space**. It is not possile to assing probability to all subsets of power set (with uncountable sample space).

Measure. A **measure** is a function

$$\mu : \mathcal{F} \to [0, \infty]$$

that satisfies:

1. $\mu(\emptyset) = 0$,

2. For any countable collection of disjoint sets $\{A_i\}$ in $\mathcal{F}$,

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

The triple $(S, \mathcal{F}, \mu)$ is called a **measure space**. Space is actualy a set with additional structure.

Types of Measures.

- **Finite measure:** $\mu(S) < \infty$.
- **Infinite measure:** $\mu(S) = \infty$.
- **Probability measure:** $\mu(S) = 1$.

Probability Measure

A **probability measure** P on $(S, \mathcal{F})$ is a function

$$P : \mathcal{F} \to [0, 1]$$

such that:

1. $P(\emptyset) = 0$,
2. $P(S) = 1$,
3. For any countable disjoint collection $\{A_i\} \subseteq \mathcal{F}$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

The summation is monotonically non-decreasing so the sum always converges.

Then $(S, \mathcal{F}, P)$ is called a **probability space**. Elements of $\mathcal{F}$ are the measurable events.

Note. $\mathcal{F}$ has smaller or equal cardinality than that of power set of S. **Basically, to study probability with nice collection of events.**

Properties of Probability Measure

For any events $A, B \in \mathcal{F}$:

1. $P(A^c) = 1 - P(A)$
2. If $A \subseteq B$, then $P(A) \leq P(B)$ (Monotonicity)
3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
4. Inclusion–Exclusion Principle. For events $A_1, \ldots, A_n$,

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i} P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) - \cdots$$

5. **Continuity of Probability.** Let $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{F}$ be events . Then

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} P(\bigcup_{n=1}^{n} A_n).$$

Proof Sketch. Define disjoint sets $B_1 = A_1$, and for $n \geq 2$ let

$$B_n = A_n \setminus A_{n-1}.$$

Then, $\{B_n\}$ are pairwise disjoint and

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n.$$

By countable additivity,

$$P\left(\bigcup_{n=1}^{m} B_n\right) = \sum_{i=1}^{m} P(B_i).$$

Taking the limit as $m \to \infty$ and using monotone convergence,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{m\to\infty} \sum_{i=1}^{m} P(B_i).$$

Thus, the probability measure P is *continuous from below*. Moving the limit outside the measure illustrates the continuity property of P.

6. Continuity of Probability (corollary). If $\{A_n\}$ is an increasing sequence of events ($A_1 \subseteq A_2 \subseteq \cdots$), then

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} P(A_n).$$

7. If $\{A_n\}$ is decreasing, then

$$P\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} P(A_n).$$

8. Union Bound:

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} P(A_n).$$

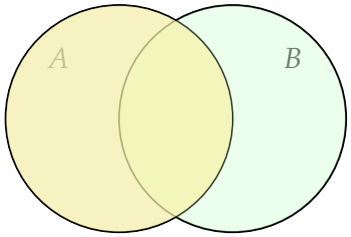


Illustration of $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Discrete Probability Space

The discrete space is easy to study since the power set of S contains all measurable sets. Here, S is countable (finite or countably infinite). Now, we can take set of all events $\mathcal{F}$ as power set $2^{|S|}$ and assign probability to each event.

If the sample space S is finite or countable, and probabilities are assigned to each outcome w_i such that

$$P(w_i) \geq 0, \quad \sum_i P(w_i) = 1,$$

then for any event $A \subseteq S$, $P(A) = \sum_{w_i \in A} P(\{w_i\})$. And, $\sum_{w_i \in S} P(w_i) = 1$.

Probabilities are actually assigned to the elements of $\mathcal{F}$ and not to the elements of S. However, for discrete case it will be like assigning probability to singleton set $w \in \mathcal{F}$. So, $P(w_i)$ is actually $P(\{w_i\})$

How is probability assigned to a singleton set? Since the theory does not prescribe a specific measure for individual singletons, we must determine it empirically through experimentation. Example of assignment if $S = \mathbb{N}$, then $p = 1/2^k$ for element k is valid measure (geometric measure). Other measures: Poisson measure.

Uncountable S

Consider the sample space $S = [0, 1]$, where we imagine throwing a dart uniformly at random onto the line segment. The dart is equally likely to land anywhere between 0 and 1, and we wish to assign a **uniform probability measure** on this interval.

However, we cannot assign probabilities to every element of the power set 2^S. The elementary idea of assigning probabilities to singletons fails in this case: there are uncountably many singleton sets, and summing their probabilities would diverge unless each singleton is assigned probability zero. Moreover, in this continuous setting, we cannot obtain probabilities of intervals simply by adding those of individual points, since the union involved is uncountable.

The key idea is therefore to **abandon the assignment of probabilities to singleton sets** and instead assign probabilities only to more meaningful subsets, such as intervals. The challenge then becomes identifying which subsets should be considered *measurable* and how to assign probabilities to them.

We would like a measure μ that satisfies the following properties:

1. (i) $\mu((a, b)) = \mu([a, b])$, i.e., open and closed intervals have the same measure.
2. (ii) **Translation invariance:** $\mu(A) = \mu(A + c)$ for any constant shift c, where $A + c = \{x + c : x \in A\}$.

Theorem (Impossibility Theorem). There does not exist a measure $\mu(A)$ defined on the full power set 2^S (that is, on all subsets of $[0, 1]$) satisfying both properties (i) and (ii) for a uniform measure.

Hence, we must **restrict** ourselves to a smaller collection of subsets—those that are still rich enough to include all "interesting" sets like intervals but small enough to admit a consistent measure. This leads to the introduction of the **Borel σ-algebra**, whose elements, called

Borel sets, are generated by open intervals through countable unions, intersections, and complements.

1.1.4 Generated and Borel σ-Algebra

We begin by asking: what should be the smallest σ-algebra on the real line $\mathbb{R}$? We would like to construct a σ-algebra that contains all intervals and is closed under complementation, countable unions, and intersections.

Generated σ-Algebra. Let C be an arbitrary collection of subsets of a sample space S (not necessarily a σ-algebra). We aim to construct the smallest σ-algebra that contains all elements of C.

Definition 1.1 (Generated σ-Algebra)**.** The *σ-algebra generated by C*, denoted $\sigma(C)$, is defined as the intersection of all σ-algebras that contain C:

$$\sigma(C) = \bigcap_{\mathcal{F} \supseteq C} \mathcal{F},$$

where the intersection is taken over all σ-algebras $\mathcal{F}$ containing C.

Theorem 1.2. There exists a smallest σ-algebra that contains all elements of C.

Proof. Let $\mathfrak{S}$ be the collection of all σ-algebras that contain C. Since the power set 2^S is itself a σ-algebra containing C, the collection $\mathfrak{S}$ is nonempty. Define

$$\sigma(C) = \bigcap_{\mathcal{F} \in \mathfrak{S}} \mathcal{F}.$$

The intersection of any family of σ-algebras is itself a σ-algebra, hence $\sigma(C)$ is a σ-algebra that contains C. By construction, it is the smallest such σ-algebra. □

Borel σ-Algebra. If C is chosen to be the collection of all open intervals in $\mathbb{R}$, then the generated σ-algebra

$$\mathcal{B}(\mathbb{R}) = \sigma(C)$$

is called the *Borel σ-algebra*. It is the smallest σ-algebra that contains all open intervals of $\mathbb{R}$. The sets belonging to $\mathcal{B}(\mathbb{R})$ are called *Borel sets*.

Proposition 1.3. Let $b \in (0, 1]$. Then the singleton set $\{b\}$ is a Borel set.

Proof Sketch. We can express the singleton $\{b\}$ as a countable intersection of open intervals:

$$\{b\} = \bigcap_{n=1}^{\infty} (b - \tfrac{1}{n},\ b + \tfrac{1}{n}).$$

Since each interval $(b - \frac{1}{n},\ b + \frac{1}{n})$ is an open set, it belongs to the Borel σ-algebra. Because the Borel σ-algebra is closed under countable intersections, it follows that $\{b\}$ is a Borel set. □

Examples of Non-Borel Sets. While all open, closed, and countable sets are Borel, there also exist subsets of $\mathbb{R}$ that are *not* Borel measurable. Famous examples include:

- **Vitali sets** — constructed using the Axiom of Choice by partitioning $\mathbb{R}$ into equivalence classes modulo the rationals.
- **Banach–Tarski paradoxical sets** — subsets of $\mathbb{R}^3$ used in the Banach–Tarski paradox, which demonstrates that certain sets can be decomposed and reassembled into a copy of themselves.

These examples highlight the existence of highly non-constructive sets lying outside the Borel σ-algebra.

Borel Sets and Lebesgue Measure

$$S = (0, 1]$$

Consider the measure space

$$\mathcal{M} = (S, \mathcal{B}),$$

where $\mathcal{B}$ denotes the Borel σ-algebra on S. We aim to define a *uniform measure* on $\mathcal{M}$.

The intuitive idea is to use *length* as the measure for intervals. However, since Borel sets include not only intervals but also more complex (and potentially pathological) subsets, the notion of length must be extended to all elements of $\mathcal{B}$.

Let $\mathcal{F}_0$ be the collection of all subsets of S that are finite unions of disjoint half-open intervals of the form $(a, b]$, together with the null set. For each $A \in \mathcal{F}_0$, we define

$$\mu(A) = \sum_i (b_i - a_i),$$

where the $(a_i, b_i]$ are the disjoint intervals forming A. This function μ represents the intuitive "length" of the set A.

The collection $\mathcal{F}_0$ forms an *algebra* of sets, but it is not a σ-algebra. To obtain a σ-algebra, we consider

$$\sigma(\mathcal{F}_0) = \mathcal{B}((0, 1]),$$

the Borel σ-algebra generated by open (or equivalently half-open) intervals.

Using the *Carathéodory Extension Theorem*, the measure μ defined on $\mathcal{F}_0$ can be uniquely extended to a measure on $\mathcal{B}((0, 1])$. This extended measure is the *Lebesgue measure* restricted to $(0, 1]$, and it coincides with our intuitive notion of length on all Borel subsets.

For example, for any singleton $\{b\} \subseteq (0, 1]$, we have

$$P(\{b\}) = 0,$$

and for the set of rationals $Q \cap (0, 1]$, we also have

$$P(Q \cap (0, 1]) = 0.$$

More generally, the Lebesgue measure on $\mathbb{R}$ is denoted by

$$(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda),$$

where λ is infinite and satisfies

$$\lambda((a, b)) = b - a,$$

representing the standard notion of length on intervals.

1.2 Linear Algebra

This section is under construction.

1.3 Calculus

This section is under construction.

1.4 Information Theory

This section is under construction.

2

Policy Gradient

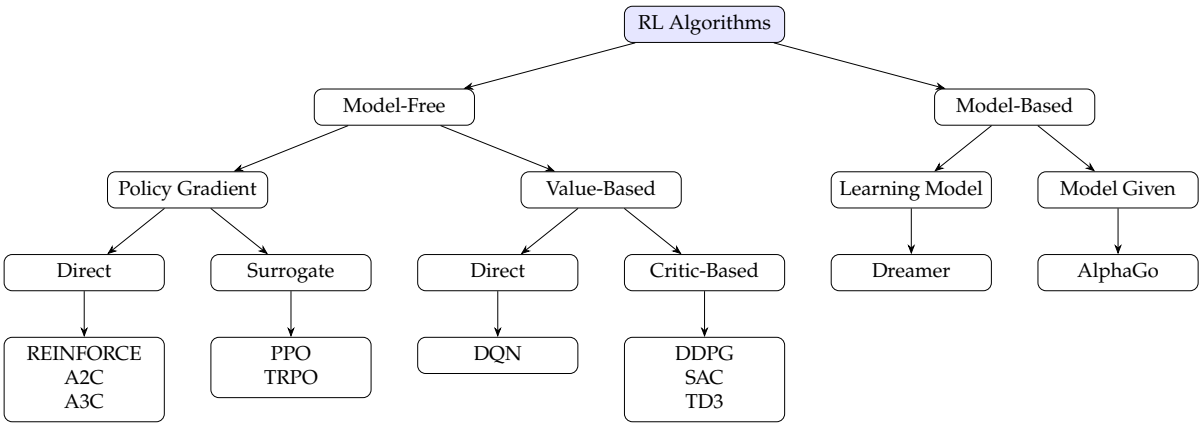


Figure 2.1: Hierarchy of RL algorithms with grouped methods.

2.1 Policy Optimization with REINFORCE

Setup. Let $\pi_\theta(a \mid s)$ be a differentiable stochastic policy with parameters θ. For a trajectory $\tau = (s_0, a_0, \ldots, s_T)$, define the (discounted) return

$$R(\tau) \;=\; \sum_{t=0}^{T-1} \gamma^t \, r_t, \qquad \gamma \in [0,1].$$

The performance objective is the expected return

$$J(\theta) \;=\; \mathbb{E}_{\tau \sim \pi_\theta}\big[\, R(\tau) \,\big].$$

Trajectory likelihood and log-derivative trick. Under a fixed environment with initial-state distribution ρ_0 and dynamics P (both independent of θ),

$$P(\tau \mid \theta) \;=\; \rho_0(s_0) \prod_{t=0}^{T-1} P(s_{t+1} \mid s_t, a_t)\, \pi_\theta(a_t \mid s_t), \quad \log P(\tau \mid \theta) \;=\; \text{const} + \sum_{t=0}^{T-1} \log \pi_\theta(a_t \mid s_t).$$

Using $\nabla_\theta P = P\,\nabla_\theta \log P$,

$$\nabla_\theta J(\theta) = \nabla_\theta \int P(\tau \mid \theta)\, R(\tau)\, d\tau = \int P(\tau \mid \theta)\, \nabla_\theta \log P(\tau \mid \theta)\, R(\tau)\, d\tau$$
$$= \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t \mid s_t)\; R(\tau) \right].$$

Expected Grad-Log-Prob (EGLP) lemma. For any distribution P_θ with density differentiable in θ,

$$\mathbb{E}_{x\sim P_\theta}\big[\nabla_\theta \log P_\theta(x)\big] \;=\; 0.$$

This follows from $\int \nabla_\theta P_\theta(x)\, dx = \nabla_\theta 1 = 0$.

Reward-to-go form ("don't let the past distract you"). It is preferable to attribute credit to an action only through *subsequent* rewards. Writing $\hat{R}_t \coloneqq \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}$,

$$\nabla_\theta J(\theta) \;=\; \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t \mid s_t)\; \hat{R}_t \right],$$

which attains lower variance than using $R(\tau)$, since terms involving past rewards cancel in expectation but add noise.

Baselines and advantages. By EGLP, for any function $b : \mathcal{S} \to \mathbb{R}$,

$$\mathbb{E}_{a_t\sim\pi_\theta(\cdot\mid s_t)}\big[\nabla_\theta \log \pi_\theta(a_t \mid s_t)\, b(s_t)\big] \;=\; 0,$$

so subtracting a baseline does not change the expectation:

$$\nabla_\theta J(\theta) \;=\; \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t \mid s_t)\, \big(\hat{R}_t - b(s_t)\big) \right].$$

The variance-minimizing choice (within state-only baselines) is $b(s_t) = V^\pi(s_t)$, yielding the *advantage* $A^\pi(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t)$ and the canonical form

$$\nabla_\theta J(\theta) \;=\; \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t \mid s_t)\, A^\pi(s_t, a_t) \right].$$

In practice, A^π is estimated from data, e.g., via Monte Carlo $\hat{R}_t - \hat{V}(s_t)$ or GAE.

Sample estimator (stochastic gradient). Given a batch $\mathcal{D}$ of trajectories sampled under the current policy, with per-time-step weights $w_t \in \{\hat{R}_t,\ \hat{A}_t\}$,

$$\widehat{\nabla_\theta J}(\theta) \;=\; \frac{1}{|\mathcal{D}|} \sum_{\tau\in\mathcal{D}} \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t \mid s_t)\; w_t.$$

Algorithm 2.1 Reinforce with Reward-to-Go and Baseline

Input: policy π_θ, critic $\hat{V}_\phi$, horizon T, discount γ, learning rate α
repeat
  Collect N trajectories by rolling out π_θ; store $(s_t, a_t, r_t)_{t=0}^{T-1}$
  for each trajectory **do**
    Compute returns-to-go $\hat{R}_t \leftarrow \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}$
    (Optional) Estimate advantages $\hat{A}_t \leftarrow \hat{R}_t - \hat{V}_\phi(s_t)$ (or GAE)
  end for
  Form policy loss $L(\theta) \leftarrow -\frac{1}{|\mathcal{D}|} \sum_{\tau,t} \log \pi_\theta(a_t \mid s_t)\, w_t$
  Update policy: $\theta \leftarrow \theta - \alpha \nabla_\theta L(\theta)$
  (Optional) Update critic ϕ by regression to targets $\hat{R}_t$
until converged

2.1.1 Generalized Advantage Estimation

Policy gradient methods rely on the notion of the *advantage function*, defined as

$$A^\pi(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t), \tag{2.1}$$

which measures how much better (or worse) an action a_t is compared to the average behavior at state s_t.

2.1.2 From Advantage to Temporal-Difference Errors

Using the Bellman equation for Q^π, we have

$$Q^\pi(s_t, a_t) = \mathbb{E}\big[r_t + \gamma V^\pi(s_{t+1}) \mid s_t, a_t\big]. \tag{2.2}$$

Substituting into the definition of the advantage,

$$A^\pi(s_t, a_t) = \mathbb{E}\big[r_t + \gamma V^\pi(s_{t+1}) - V^\pi(s_t) \mid s_t, a_t\big]. \tag{2.3}$$

We define the *temporal-difference (TD) error* at time t as

$$\delta_t = r_t + \gamma V^\pi(s_{t+1}) - V^\pi(s_t). \tag{2.4}$$

Thus, the advantage can be written simply as the expected TD error:

$$A^\pi(s_t, a_t) = \mathbb{E}[\delta_t \mid s_t, a_t]. \tag{2.5}$$

2.1.3 Multi-step Advantage Estimators

Instead of relying only on the 1-step TD error, one can consider a k-step return. Summing discounted TD errors yields

$$\sum_{i=0}^{k-1} \gamma^i \delta_{t+i} = \Big(r_t + \gamma r_{t+1} + \cdots + \gamma^{k-1} r_{t+k-1} + \gamma^k V(s_{t+k})\Big) - V(s_t). \tag{2.6}$$

The telescoping structure shows that the k-step estimator corresponds to the k-step bootstrapped return minus the value baseline.

2.1.4 Generalized Advantage Estimation

Generalized Advantage Estimation (GAE) [1] combines all k-step estimators into a single exponentially-weighted average:

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = (1-\lambda)\sum_{k=1}^{\infty}\lambda^{k-1}\left(\sum_{i=0}^{k-1}\gamma^i\delta_{t+i}\right). \tag{2.7}$$

Rearranging terms yields the compact recursive form

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \delta_t + \gamma\lambda\delta_{t+1} + (\gamma\lambda)^2\delta_{t+2} + \cdots. \tag{2.8}$$

In practice, this is implemented efficiently by the backward recursion

$$\hat{A}_t = \delta_t + \gamma\lambda\hat{A}_{t+1}, \tag{2.9}$$

with $\hat{A}_T = 0$ at the end of a trajectory.

2.1.5 Bias–Variance Tradeoff

The parameter $\lambda \in [0,1]$ interpolates between high-bias, low-variance and low-bias, high-variance estimators:

- $\lambda = 0$: Uses only the 1-step TD error ($\hat{A}_t = \delta_t$), yielding low variance but higher bias.
- $\lambda = 1$: Recovers the Monte Carlo return ($\hat{A}_t = R_t - V(s_t)$), yielding low bias but higher variance.
- $0 < \lambda < 1$: Provides a balance between variance reduction and bias control, which is crucial for stable policy gradient methods such as PPO and TRPO.

2.2 Reinforce Extensions

Algorithms like A2c and A3c

2.3 Surrogate Objective

it covers natural pg, trpo and ppo

2.3.1 Surrogate Objective, Natural Gradient, and TRPO

True objective. Let $J(\pi)$ denote the (discounted) expected return of a policy π:

$$J(\pi) = \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{\infty}\gamma^t r(s_t,a_t)\right], \quad d^\pi(s) = (1-\gamma)\sum_{t=0}^{\infty}\gamma^t \Pr(s_t = s \mid \pi)$$

where d^π is the discounted state-visitation distribution.

Performance difference and the role of advantages. For any two policies π and π_{old}, the performance difference lemma gives

$$J(\pi) \;=\; J(\pi_{\text{old}}) \;+\; \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d^{\pi},\,a\sim\pi(\cdot|s)}\Big[\,A^{\pi_{\text{old}}}(s,a)\,\Big], \tag{2.10}$$

where $A^{\pi_{\text{old}}}(s,a) = Q^{\pi_{\text{old}}}(s,a) - V^{\pi_{\text{old}}}(s)$ is the advantage under π_{old}. Eq. (2.10) is exact, but involves the *unknown* distribution d^{π} induced by the *new* policy.

From J to a surrogate L: switching distributions. We would like to evaluate/improve π using samples from π_{old}. There are two distributions in (2.10): (i) actions $a \sim \pi(\cdot|s)$ and (ii) states $s \sim d^{\pi}$.

Action distribution. For the inner expectation we use a likelihood-ratio identity:

$$\mathbb{E}_{a\sim\pi(\cdot|s)}[f(s,a)] \;=\; \mathbb{E}_{a\sim\pi_{\text{old}}(\cdot|s)}\left[\frac{\pi(a|s)}{\pi_{\text{old}}(a|s)}\,f(s,a)\right].$$

State distribution. Replacing d^{π} by $d^{\pi_{\text{old}}}$ yields a tractable, *local* approximation. This replacement is what makes the objective a *surrogate*: at $\pi = \pi_{\text{old}}$ it matches J to first order (i.e., it has the same gradient), while being computable from data gathered under π_{old}.

Putting the two pieces together, the standard surrogate objective around π_{old} is

$$\begin{aligned} L_{\pi_{\text{old}}}(\pi) &:= J(\pi_{\text{old}}) \;+\; \frac{1}{1-\gamma}\,\mathbb{E}_{\substack{s\sim d^{\pi_{\text{old}}}\\ a\sim\pi_{\text{old}}(\cdot|s)}}\Big[\,r_{\pi/\pi_{\text{old}}}(s,a)\,A^{\pi_{\text{old}}}(s,a)\,\Big], \\ &\text{with}\quad r_{\pi/\pi_{\text{old}}}(s,a) := \frac{\pi(a|s)}{\pi_{\text{old}}(a|s)}. \end{aligned} \tag{2.11}$$

(One often drops the constant $J(\pi_{\text{old}})$ when optimizing, since it does not depend on π.)

First-order approximation of the surrogate. Recall the surrogate objective around π_{old}:

$$L_{\pi_{\text{old}}}(\pi_\theta) = J(\pi_{\text{old}}) + \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d^{\pi_{\text{old}}},\,a\sim\pi_\theta(\cdot|s)}[A^{\pi_{\text{old}}}(s,a)]\,.$$

Let θ_{old} be the parameters of the current policy and $\Delta\theta = \theta - \theta_{\text{old}}$ the update. Performing a first-order Taylor expansion around θ_{old} gives

$$L_{\pi_{\text{old}}}(\pi_\theta) \;\approx\; L_{\pi_{\text{old}}}(\pi_{\theta_{\text{old}}}) + g^{\top}\Delta\theta,$$

where

$$g \;=\; \nabla_\theta L_{\pi_{\text{old}}}(\pi_\theta)\big|_{\theta=\theta_{\text{old}}} = \mathbb{E}_{s,a\sim\pi_{\text{old}}}\left[\nabla_\theta \log \pi_\theta(a|s)\,A^{\pi_{\text{old}}}(s,a)\right].$$

Thus, to first order, maximizing L reduces to taking a step in the policy gradient direction. This shows that the surrogate objective recovers the REINFORCE gradient at $\pi = \pi_{\text{old}}$, while also providing a local approximation of $J(\pi)$ that can be combined with additional constraints such as a KL trust region.

Why it is a surrogate (first-order accuracy). Let π_θ be a differentiable policy, θ_{old} the current parameters, and $\Delta\theta = \theta - \theta_{\text{old}}$. Linearizing $L_{\pi_{\text{old}}}$ at θ_{old} gives

$$L_{\pi_{\text{old}}}(\pi_\theta) \approx L_{\pi_{\text{old}}}(\pi_{\theta_{\text{old}}}) + g^{\top}\Delta\theta, \qquad g \;=\; \nabla_\theta L_{\pi_{\text{old}}}(\pi_\theta)\big|_{\theta=\theta_{\text{old}}} = \mathbb{E}_{s,a\sim\pi_{\text{old}}}\left[\nabla_\theta \log \pi_\theta(a|s)\,A^{\pi_{\text{old}}}(s,a)\right],$$

which equals the true policy gradient $\nabla_\theta J(\pi_\theta)|_{\theta=\theta_{\text{old}}}$. Thus L matches J up to first order at θ_{old}.

Natural gradient via a local KL constraint. To control the size of the update, constrain the average KL divergence from the old policy:

$$\bar{D}_{\mathrm{KL}}(\theta_{\mathrm{old}}, \theta) \;:=\; \mathbb{E}_{s\sim d^{\pi_{\mathrm{old}}}}\left[D_{\mathrm{KL}}\big(\pi_{\theta_{\mathrm{old}}}(\cdot|s)\,\|\,\pi_\theta(\cdot|s)\big)\right] \;\le\; \delta.$$

Approximating the objective to first order and the KL to second order around θ_{old} yields

$$\max_{\Delta\theta}\; g^\top \Delta\theta \quad \text{s.t.} \quad \tfrac{1}{2}\,\Delta\theta^\top F\,\Delta\theta \;\le\; \delta, \qquad F \;=\; \mathbb{E}_{s,a\sim\pi_{\mathrm{old}}}\big[\nabla \log \pi_\theta(a|s)\,\nabla \log \pi_\theta(a|s)^\top\big]_{\theta=\theta_{\mathrm{old}}}.$$

The solution is the *natural policy gradient* step

$$\Delta\theta^\star \;=\; \sqrt{\frac{2\delta}{g^\top F^{-1} g}}\, F^{-1} g,$$

which rescales the vanilla gradient by the Fisher information, producing updates that are invariant to reparameterization.

TRPO from the surrogate. Trust Region Policy Optimization (TRPO) replaces the linearized objective with the full surrogate (2.11) and enforces the KL constraint directly:

$$\max_{\theta}\; \mathbb{E}_{s,a\sim\pi_{\mathrm{old}}}[r_\theta(s,a)\,A^{\pi_{\mathrm{old}}}(s,a)] \quad \text{s.t.} \quad \mathbb{E}_{s\sim d^{\pi_{\mathrm{old}}}}\left[D_{\mathrm{KL}}\big(\pi_{\mathrm{old}}(\cdot|s)\,\|\,\pi_\theta(\cdot|s)\big)\right] \;\le\; \delta, \tag{2.12}$$

where $r_\theta(s,a) = \pi_\theta(a|s)/\pi_{\mathrm{old}}(a|s)$. TRPO approximately solves (2.12) by computing the natural gradient direction $F^{-1}g$ (via conjugate gradients) and performing a backtracking line search that ensures the empirical KL stays within the trust region and that the surrogate improves, yielding stable monotonic progress under mild assumptions.

Where the distribution switch occurs. The surrogate construction makes two explicit substitutions relative to the true objective: (i) it *replaces* d^π by $d^{\pi_{\mathrm{old}}}$ (state distribution switch), and (ii) it *reweights* the action expectation using the likelihood ratio $r_{\pi/\pi_{\mathrm{old}}}$ (action distribution switch). These steps allow us to estimate the objective and its gradient from trajectories sampled under π_{old}, while the KL trust region controls the discrepancy induced by the state-distribution replacement.

2.3.2 Proximal Policy Optimization (PPO)

Proximal Policy Optimization (PPO) [2] is an on-policy actor–critic algorithm designed to improve the stability of policy gradient methods. It alternates between (i) collecting trajectories from the current policy and (ii) optimizing a surrogate objective over several epochs of minibatch stochastic gradient ascent.

Key ideas.

- The policy $\pi_\theta(a|s)$ (actor) and value function $V_\phi(s)$ (critic) are parameterized by neural networks.
- Trajectories of length T are collected, and returns R_t and advantages A_t are computed.

- The surrogate objective penalizes large deviations from the old policy via a clipped importance ratio:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(r_t(\theta)A_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\,A_t\right)\right],$$

  where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$.

- The critic is updated by minimizing a squared-error loss $L^{\text{VF}}(\phi) = \mathbb{E}_t\left[(V_\phi(s_t) - R_t)^2\right]$.
- Advantages A_t are estimated using Generalized Advantage Estimation (GAE):

$$A_t = \sum_{l=0}^{\infty}(\gamma\lambda)^l\delta_{t+l}, \quad \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t).$$

Algorithm 2.2 Proximal Policy Optimization (PPO, clipped variant)

Initialize policy parameters θ, value parameters ϕ
for iteration = $1, 2, \ldots$ **do**
  Collect trajectories $\{(s_t, a_t, r_t)\}_{t=0}^{T-1}$ using π_θ
  Compute returns R_t and advantages A_t (e.g., with GAE)
  Store log-probabilities $\log \pi_\theta(a_t|s_t)$ as baseline
  for epoch = $1, \ldots, K$ **do**
    Sample minibatch of transitions
    Compute probability ratio $r_t(\theta) \leftarrow \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$
    Compute clipped policy loss $L^{\text{CLIP}}(\theta)$
    Update policy by ascending $\nabla_\theta L^{\text{CLIP}}(\theta)$
    Update value function by descending $\nabla_\phi L^{\text{VF}}(\phi)$
  end for
end for

Pseudocode.

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